

Dose-Finding Plate

Highest dose of each inhibitor that still grows like the control — that dose, and only that dose, goes into the real repeat. One plate, no drug, worst-case temperature.

27 °C only

no prothioconazole

every well = 5.03 mL cells + 20 µL DMSO + 10 µL inhibitor

OD₆₀₀ 0.0005 · ~4 days

nPG: 4 doses · SHAM: 5 doses

MAP 24-well plate — fill 14, rest empty

	1	2	3	4	5	6
A	A1 CTRL EtOH	A2 CTRL EtOH	A3 nPG 0.0125 mM	A4 nPG 0.025 mM	A5 nPG 0.05 mM	A6 nPG 0.1 mM
B	B1 SHAM 0.0125 mM	B2 SHAM 0.025 mM	B3 SHAM 0.05 mM	B4 SHAM 0.1 mM	B5 SHAM 0.2 mM	B6
C	C1 BLANK nPG 0.1	C2 BLANK SHAM 0.2	C3	C4	C5	C6
D	D1	D2	D3	D4	D5	D6

 control (cells + ethanol) nPG ladder SHAM ladder cell-free blank (skip if vials tight)

Positions don't matter to the reader — keeping each ladder in one row just makes the dose-response easy to eyeball as it runs.

FILL every well the same, one thing changes

cell wells: 5.03 mL cells (OD 0.0005) + 20 µL P0 (DMSO) + 10 µL inhibitor stock

blank wells (C1–C2): sterile YMS instead of cells; same DMSO + inhibitor

The 20 μL DMSO goes in even though there's no drug — it holds solvent at 0.4%, identical to the main run, so growth is comparable. Foil / dark: nPG & SHAM are light-sensitive.

STOCKS ethanol dilutions of your masters — 10 μL /well

FINAL	NPG STOCK (FROM 50 mM)	SHAM STOCK (FROM 250 mM)
0.0125 mM	6.25 mM · 125 μL +875 EtOH	6.25 mM · 25 μL +975 EtOH
0.025 mM	12.5 mM · 250 μL +750 EtOH	12.5 mM · 50 μL +950 EtOH
0.05 mM	25 mM · 500 μL +500 EtOH	25 mM · 100 μL +900 EtOH
0.1 mM	50 mM · master neat	50 mM · 200 μL +800 EtOH
0.2 mM	– (nPG stops at 0.1)	100 mM · 400 μL +600 EtOH

Every dose is 2.5–40 $\times$ below the 0.5 mM that killed growth last run. nPG caps at 0.1 mM — the 50 mM master can't be concentrated, and 0.1 already showed an effect, so the useful range is below it.

READ the plate gives two answers

The dose

For each ladder, the **highest well that grows like A1–A2** is the working dose for the repeat. Expect nPG $\approx$ 0.05–0.1, SHAM $\approx$ 0.05–0.1.

The shape

Dips then **plateaus** $\rightarrow$ on-target AOX block (real, usable). **Keeps falling** every step $\rightarrow$ chemical toxicity, go lower.

Then the real repeat: both chemicals, both temperatures (15 & 27 $^{\circ}\text{C}$), full prothioconazole dose-response, fresh culture — with doses that passed this gate.